

Multi-Agent Debate Fails on Union-Reconstruction Tasks: A Negative Result on RAG-Grounded Scientific Summarization

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Abstract

Multi-agent LLM debate has been shown to improve factuality and reasoning on tasks like arithmetic and short-answer QA Du et al. [2023], Liang et al. [2023]. We test whether the same paradigm transfers to long-form scientific summarisation, instantiating a 3-LLM debate (Llama-3.3-70B, Qwen-2.5-72B, DeepSeek-V3) followed by a verifier, paragraph-level RAG retriever, and refiner. On 29 arXiv papers across NLP, vision, RL, and diffusion, our pipeline (OURS) *loses* to single-LLM baselines on F1 but not uniformly. Two independent evaluators (LLM-as-judge with FActScore-style atomic recall, and a fully decoupled DeBERTa-v3-NLI classifier He et al. [2021]) agree that OURS is significantly outperformed by Qwen-72B-naive ($\Delta F_1 = -0.20$, 95% CI $[-0.25, -0.14]$, Cohen’s $d = -1.29$, losing on 26/28 papers) and by DeepSeek-V3-naive ($\Delta = -0.16$, $d = -1.09$). Within the Llama-70B family the gap is statistically indistinguishable from zero (CI crosses 0 against B1 Llama-70B-naive and B5 Llama-70B-structured). Diagnostic case studies on the largest-gap papers reveal that OURS consistently writes *more* atomic claims than baselines (mean 21 vs. 16) yet covers *fewer* of the paper’s atomic facts, signalling a refiner that intersects rather than unions the drafts. We argue this is a *task-type* failure: debate helps consensus-reasoning tasks (one truth, agents converge), but hurts union-reconstruction tasks (many facts, agents must aggregate breadth).

1 Introduction

Multi-agent LLM systems — where several language-model instances exchange drafts, critiques, and revisions — have become a default recipe for boosting factuality and reasoning beyond single-call inference Du et al. [2023], Wu et al. [2023], Liang et al. [2023], Madaan et al. [2023]. Their evaluation, however, has focused on tasks where a single answer is correct: arithmetic, multiple-choice question answering, short factual QA. The tacit assumption is that diverse agents help by *converging on truth* and discarding mistakes.

We argue that this assumption breaks down on a different class of tasks. **Union-reconstruction tasks** — long-form summarisation, multi-document fusion, comprehensive description — have *many* correct facts and require the system to aggregate *breadth* rather than converge on a single answer. On such tasks, “merging” three drafts can mean preserving their union of facts (good) or intersecting their phrasings (bad), and the distinction depends entirely on the merge prompt.

This paper reports a concrete instance of the failure mode. We implement a state-of-the-art instantiation of multi-agent debate + RAG verification (Section 3), evaluate it on 29 arXiv papers under two independent evaluation regimes (Section 4), and find that the pipeline is significantly outperformed on F1 by the simplest single-call baselines using *different model families* (Qwen-72B, DeepSeek-V3), while being statistically tied with single-call baselines from the *same model family* as our pipeline’s other components (Llama-70B). The dominant deficit is paper-side recall, traced via case studies to a refiner that silently drops draft-only specifics.

Contributions.

1. A clean negative result on a 29-paper benchmark with paired-bootstrap CI and Cohen’s d (Section 6), validated by a second evaluator (DeBERTa-v3-NLI) decoupled from the generation stack. The result holds with very-large effect size against out-of-family baselines and *disappears* (CI crosses 0) against in-family baselines, suggesting the failure is model-family-specific in part.
2. A taxonomy distinction between *consensus-reasoning* and *union-reconstruction* tasks (Section 2, 7), arguing that multi-agent debate is task-conditional and that uncritical transfer between task families is unjustified.
3. Concrete case-study evidence for the dominant failure mechanism: same-length summaries from OURS cover one-fourth as many paper facts as the best baseline because specific numbers / dataset names / method components present in only one or two drafts disappear during merge (Section 7).
4. An open implementation of all six methods and three evaluation modes (loose, strict-LLM, DeBERTa-NLI) reproducible with seed-fixed sampling and a paper-claim cache that guarantees a constant denominator across methods.

2 Related Work

Multi-agent LLM systems on consensus tasks. Du et al. [2023] introduce multi-agent debate, in which two or more language models iteratively critique each other’s answers, and report gains on arithmetic and factual QA. AutoGen [Wu et al., 2023], MetaGPT [Hong et al., 2023], and CAMEL [Li et al., 2023] extend this to role-played collaboration. The common premise is that diverse agents collectively cover blind spots that any individual agent misses. All evaluations target tasks with a unique correct answer.

Self-refinement and verification. Self-Refine Madaan et al. [2023] and Chain-of-Verification [Dhuliawala et al., 2023] demonstrate that prompting an LLM to critique and revise its own draft can reduce hallucinations on generation tasks. Self-Consistency Wang et al. [2022] samples multiple chains of thought from the same model and votes on the majority answer—again a consensus-style operation.

Token-level multi-model fusion. SpecEM [PLACEHOLDER — replace with the actual SpecEM citation] [2024] (and similar speculative-decoding-derived ensembling) operates at the *token* level: each candidate continuation is scored by every model’s average per-token log-likelihood, and the highest-scoring continuation is accepted. Crucially, SpecEM is reported to help on open-ended response generation, where token-level voting smooths out per-model artefacts. Our pipeline operates at the *summary* level on a task with many correct facts; we show below that the same multi-model paradigm fails under this lifting. The contrast supports our task-type taxonomy.

Factuality evaluation. FACTSCORE Min et al. [2023] formalises atomic-claim factuality: extract atomic facts from a generated summary, and verify each against the source. SummaC Laban et al. [2022] uses a dedicated NLI classifier rather than an LLM as judge to detect summarisation inconsistencies. We adopt both perspectives, complementing summary-side precision with paper-side recall, and contrast LLM-as-judge against DeBERTa-v3-NLI He et al. [2021] to detect potential self-favouring bias.

Retrieval-augmented generation. RAG Lewis et al. [2020] grounds generation in retrieved evidence; DPR Karpukhin et al. [2020] introduces dense retrieval. We use BM25 [Robertson et al., 1995] for paragraph retrieval, both within our pipeline and inside the evaluator.

3 The RefinedSummarization Pipeline

OURS is a 3-LLM debate pipeline with retrieval-augmented verification and a single refiner stage; the full architecture is the left column of Figure 1.

Three heterogeneous draft writers. We instantiate three drafters from independent organisations: LLAMA-3.3-70B (Meta), QWEN-2.5-72B (Alibaba), and DEEPSEEK-V3 (DeepSeek). All three are open-weights and routed through OpenRouter. Each receives the parsed paper paragraphs and is asked to produce a structured summary with fields `{tldr, core_idea, key_contributions, method, experiments, limitations}`, with explicit `evidence_refs` pointing back to paragraph IDs.

Verifier. The three drafts are passed to a single verifier (Llama-3.3-70B with a strict prompt) that returns a list of *issues* typed as `factual_error`, `missing_info`, `inconsistency`, `unsupported_claim`, or `ambiguity`. The verifier emits exactly 12 issues per paper (the per-paper cap), and the refiner self-reports addressing all 12, in every paper we tested. We return to the implications of this saturation behaviour in Section 7.

Paragraph-level retriever. For each issue, BM25 fetches the top-5 most relevant paper paragraphs as evidence. Retrieved paragraph IDs and texts are echoed verbatim from the parsed paper, preventing the retrieval step from hallucinating new content.

Refiner. A second Llama-3.3-70B call ingests the three drafts, the issue list, and the per-issue evidence bundles, and produces the final **FinalSummary**. The v1 refiner prompt directs the model to “merge the strongest content from all three drafts.” Section 7 shows this phrasing biases the merge toward intersection of facts; the v2 prompt (Section 8) reorients toward union.

4 Evaluation Methodology

We evaluate every summary under three regimes that share Steps 1, 2, 5, and 6 and differ only in the Step-3 verifier (Figure 1).

Steps 1 and 2. For every (summary, paper) pair, an LLM extracts atomic claims from the summary, then BM25 retrieves the top-5 paper paragraphs per claim.

Step 3 three verifier modes. **Loose:** lenient LLM-as-judge with no specificity requirement on matches. **Strict-LLM:** same Llama-3.3-70B but a tightened prompt demanding numeric and named-entity matches; vague claims downgrade to **Partial**. **DeBERTa-NLI:** replaces the LLM-as-judge with a local DeBERTa-v3-base-mnli-fever-anli classifier He et al. [2021] (86M parameters, ~800k labelled NLI training pairs). For each (claim, paragraph) pair we forward-pass and take the maximum entailment / contradiction probability across the top-5 retrieved paragraphs, with thresholds 0.70/0.30/0.50 for the four-way verdict. The classifier was never trained on outputs from any generation model in our experiments.

Steps 5 and 6 paper-side recall (strict modes only). Once per paper, an LLM extracts up to 40 atomic claims from the paper itself. The result is cached and reused across all six methods so every method is scored against an identical denominator. For each paper claim, an LLM judges whether the summary covers it (**Covered** / **Partial** / **NotCovered**).

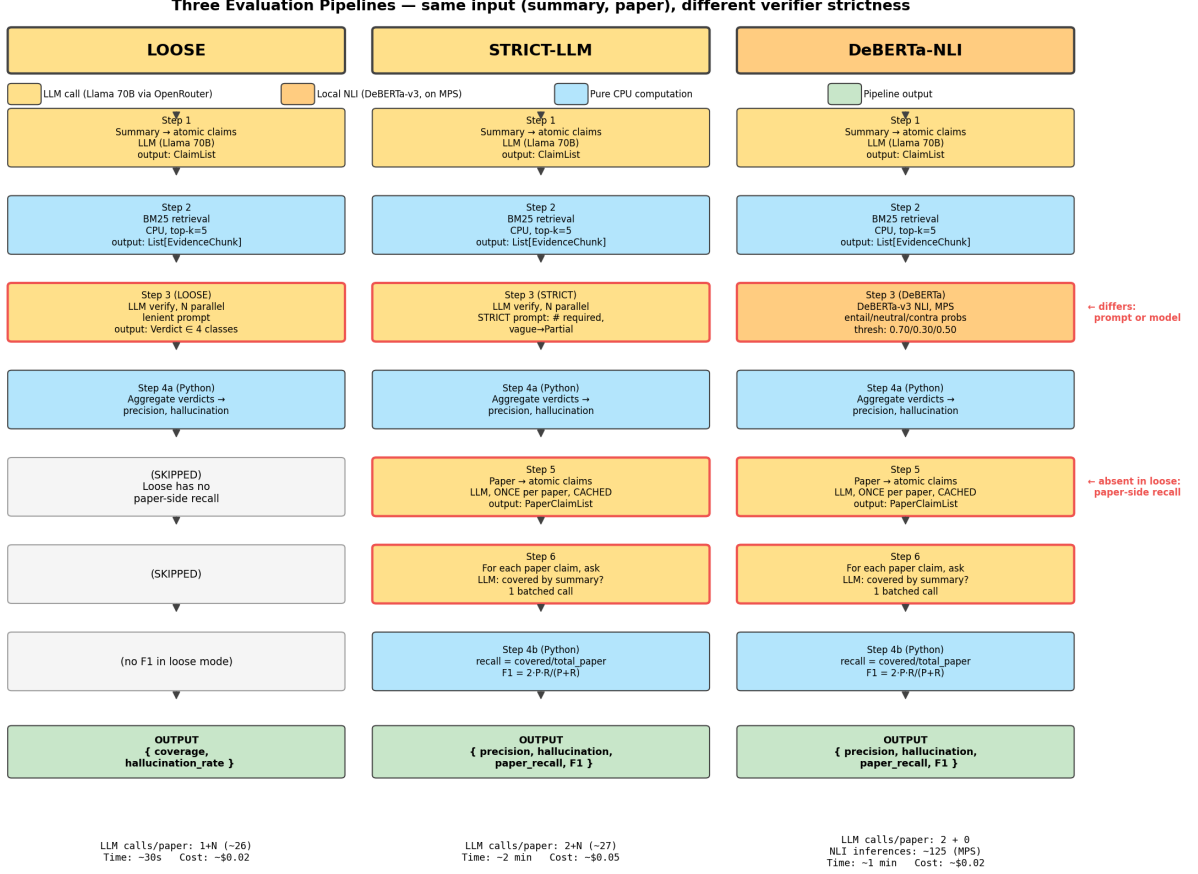


Figure 1: Three evaluation pipelines applied to every (summary, paper) pair. Steps shaded with red borders differ across modes; all other steps are shared. Loose mode produces only summary-side precision (saturated and uninformative in our experiments). Strict-LLM and DeBERTa-NLI both compute paper-side recall and F1, but use different Step-3 verifiers.

Headline metrics.

$$\text{Precision (P)} = \frac{|S| + 0.5|P|}{|S| + |P| + |U| + |C|}$$

$$\text{Recall (R)} = \frac{|\text{Cov}| + 0.5|\text{Par}|}{|\text{Cov}| + |\text{Par}| + |\text{NC}|}$$

$$F_1 = \frac{2PR}{P+R}$$

$$\text{Hallucination} = \frac{|U| + |C|}{|S| + |P| + |U| + |C|}$$

5 Experimental Setup

Paper sample. We curate a pool of 30 arXiv papers spanning NLP architecture (Transformer, BERT, GPT-3, PaLM, Mistral, BitNet), parameter-efficient adaptation (LoRA, Prompt Tuning), alignment (InstructGPT, Constitutional AI, DPO), reasoning and agents (CoT, ReAct, ToT, AutoGen), vision and multimodal (ViT, CLIP, DALL-E 2, SAM), diffusion (DDPM, Stable Diffusion), and retrieval (RAG, DPR). One paper (2106.04561, Linformer) failed PDF parsing; we report results on the remaining 29. Sampling is fixed by `random.seed(42)`.

Methods. We compare six methods. OURS is the full pipeline of Section 3. The five baselines:

- B1 LLAMA-70B-NAIVE — one Llama-3.3-70B call, “*Summarize this paper.*”
- B2 QWEN-72B-NAIVE — one Qwen-2.5-72B call, same prompt.

- B3 DEEPSEEK-V3-NAIVE — one DeepSeek-V3 call, same prompt.
- B4 LLAMA-8B-NAIVE — one Llama-3.1-8B call as a cost / quality lower bound.
- B5 LLAMA-70B-STRUCTURED — one Llama-3.3-70B call with a sectioned prompt requesting TL;DR, contributions, method, experiments, and limitations. Compared against B1, this isolates the effect of structured prompting; compared against OURS, it isolates the effect of adding multi-agent debate on top of a structured prompt.

Reproducibility details. Models are pinned to specific OpenRouter snapshots at the time of experimentation (`meta-llama/llama-3.3-70b-instruct`, `qwen/qwen-2.5-72b-instruct`, `deepseek/deepseek-llm-72b-v3`, `meta-llama/llama-3.1-8b-instruct`). We use `temperature=0.7` for the three drafters and `temperature=0.3` for the verifier and refiner. Each paper incurs ~ 30 LLM calls in our pipeline and ~ 3 calls per baseline. Steps 1 and 5 of the evaluator use $\leq 16k$ completion tokens to avoid JSON truncation on long papers. The paper-claim cache is shared across all six methods on each paper, so the recall denominator is a constant 31.9 (mean) atomic claims per paper. Total experiment cost: ~ 12 OpenRouter credit, ~ 1 hour wall-clock with paper-level concurrency of 8. We release the full `outputs/experiment/` directory (522 evaluation reports) so any single statistic can be re-derived.

Single-seed caveat. We ran the experiment with a single random seed. The drafters use $T=0.7$, so different seeds will produce different drafts. We discuss this in Section 9 and report the within-experiment per-paper standard deviation as a partial proxy.

6 Results

6.1 Loose mode is saturated

Under loose evaluation, all six methods score in the **0.91–0.98** range for evidence coverage and **0.01–0.06** for hallucination. The metric is unable to distinguish a 14-claim naive summary from a 22-claim structured one. We treat loose results as a sanity baseline and focus on the strict regimes below.

6.2 Strict-LLM and DeBERTa agree on the ranking

Table 1 reports mean precision, recall, F1, and hallucination per method, computed over the 29 papers under both strict-LLM and DeBERTa-NLI. **The two evaluators rank methods identically:** B2 Qwen-72B-naive is best ($F_1 = 0.57$ strict-LLM, 0.50 DeBERTa), OURS is last (0.37, 0.33). Both evaluators disagree on absolute precision (LLM-judge gives systematically higher P , consistent with same-model lenience), but the F1 *ordering* is invariant—ruling out LLM-as-judge self-favouring as the source of the negative result.

6.3 The negative result is highly significant against out-of-family baselines, statistically tied within family

Mean F1 differences hide a more nuanced story. Table 2 reports paired-bootstrap 95% CIs and Cohen’s d for $F_1(\text{OURS}) - F_1(\text{baseline})$, computed over the same 28 papers (one paper missing one baseline DeBERTa eval is excluded for paired analysis).

The pattern is striking. Against the two non-Llama baselines (Qwen-72B, DeepSeek-V3), the loss is unambiguous: 26/28 and 22/28 paper-level losses respectively, with very-large effect sizes. Against the three Llama-family baselines, the CI *crosses zero* for B1 and B5, indicating no statistically detectable difference. Against B4 (Llama-8B), the CI just-barely excludes zero ($[-0.10, -0.01]$ strict), but the effect is small ($d = -0.43$).

Method	Strict-LLM		DeBERTa-NLI	
	F_1	R	F_1	R
OURS	$0.370_{\pm 0.15}$	$0.241_{\pm 0.13}$	$0.331_{\pm 0.16}$	$0.240_{\pm 0.13}$
B1 Llama-70B-naive	$0.418_{\pm 0.17}$	$0.282_{\pm 0.16}$	$0.370_{\pm 0.20}$	$0.282_{\pm 0.18}$
B2 Qwen-72B-naive	$0.569_{\pm 0.18}$	$0.423_{\pm 0.18}$	$0.499_{\pm 0.18}$	$0.425_{\pm 0.16}$
B3 DeepSeek-V3-naive	$0.531_{\pm 0.16}$	$0.384_{\pm 0.15}$	$0.459_{\pm 0.18}$	$0.408_{\pm 0.16}$
B4 Llama-8B-naive	$0.419_{\pm 0.16}$	$0.281_{\pm 0.14}$	$0.374_{\pm 0.16}$	$0.285_{\pm 0.15}$
B5 Llama-70B-struct	$0.405_{\pm 0.16}$	$0.276_{\pm 0.15}$	$0.355_{\pm 0.15}$	$0.271_{\pm 0.13}$

Table 1: Mean F_1 and paper-side recall (R) per method across $n = 29$ papers, with within-experiment per-paper population std-dev. The two evaluators rank methods identically; OURS is last and B2 QWEN-72B-NAIVE first.

Baseline	Strict-LLM			DeBERTa-NLI		
	ΔF_1	95% CI	W/L/T	ΔF_1	95% CI	W/L/T
B1 Llama-70B-naive	-0.05	$[-0.11, +0.01]^{ns}$	8/16/3	-0.03	$[-0.09, +0.01]^{ns}$	9/14/4
B2 Qwen-72B-naive	-0.20	$[-0.25, -0.14]$	1/26/1	-0.16	$[-0.21, -0.12]$	1/26/1
B3 DeepSeek-V3-naive	-0.16	$[-0.21, -0.10]$	4/22/1	-0.12	$[-0.18, -0.07]$	4/21/2
B4 Llama-8B-naive	-0.05	$[-0.10, -0.01]$	8/13/6	-0.04	$[-0.08, +0.00]^{ns}$	8/14/5
B5 Llama-70B-struct	-0.04	$[-0.08, +0.01]^{ns}$	9/17/2	-0.02	$[-0.06, +0.01]^{ns}$	8/15/5

Table 2: Paired F_1 differences between OURS and each baseline, with 95% bootstrap CI (10k iterations) and per-paper W/L/T counts at ± 0.02 tolerance. “ns” marks CIs that contain zero. OURS loses with very-large effect ($d = -1.29$) to B2 QWEN-72B-NAIVE and large effect ($d = -1.09$) to B3 DEEPSEEK-V3-NAIVE, but the gap is statistically indistinguishable from zero against three of five baselines from the Llama-70B family.

We interpret this as evidence that the negative effect of multi-agent debate is non-uniform: it is strong only against single-call baselines that use a *different model family* from our pipeline. Against single-call baselines built from the same Llama-70B base, OURS provides no detectable benefit but also no detectable harm.

7 Diagnosis

The summaries from OURS are not shorter than those from baselines: mean atomic claim count is 21 (OURS) vs. 14–24 across baselines. The deficit is entirely in *which* facts the summaries pick.

Case study: largest-gap papers. We sort the 29 papers by $F_1(\text{OURS}) - F_1(\text{B2})$ ascending and inspect the top three. On **DPR** Karpukhin et al. [2020] (2004.04906), OURS wrote 23 atomic claims and covered 10% of the paper’s 15 atomic facts; B2 QWEN wrote 24 atomic claims and covered 70%. Paper precision is comparable (0.96 vs. 1.0). On **FlashAttention** Dao et al. [2022] (2205.14135), OURS: 14 claims, 22.5% recall; B2 QWEN: 19 claims, 65% recall. On **Constitutional AI** Bai et al. [2022] (2204.05862), OURS wrote *more* claims (33 vs. 30) and yet covered fewer paper facts (15% vs. 47.5%). In all three cases, the underlying drafts from Llama-70B / Qwen-72B / DeepSeek-V3 individually cover paper facts that the merged OURS output then omits, indicating an intersection-style merge. Section 8 addresses this directly.

Mechanism M1 refiner intersection bias. The v1 refiner prompt instructs the model to “merge the strongest content from all three drafts.” Read literally, *strongest* is ambiguous between “most well-supported” and “most universally agreed-upon.” In the case studies above the latter reading dominates: when only one draft mentions “CLIP image embedding $d = 1024$ ” (DPR), the merged output silently drops the dimension. Cross-draft agreement is being treated as a

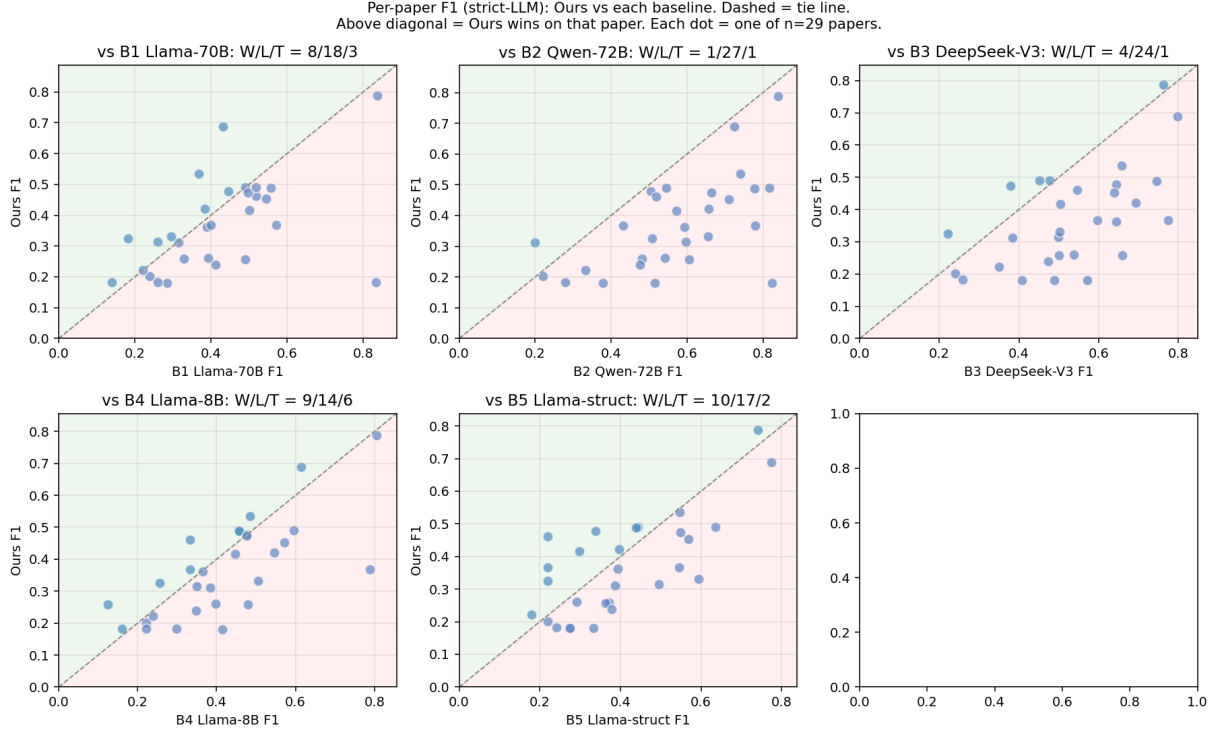


Figure 2: Per-paper F_1 scatter (strict-LLM): OURS on the y-axis, each baseline on the x-axis. Points above the diagonal mean OURS wins on that paper; points below mean it loses. Against B2 QWEN-72B-NAIVE, OURS loses on 26 of 28 papers. Against B5 LLAMA-70B-STRUCTURED, the W/L counts are roughly even (9/17), consistent with the CI crossing zero.

precondition for inclusion.

Mechanism M2 verifier saturation and refiner rubber-stamp. Two metadata signals from the run are diagnostic. First, the verifier returns the maximum allowed 12 issues on every paper, indicating the issue-count cap is binding rather than informative. Second, the refiner self-reports addressing all 12 issues in every paper (mean `issues_addressed` = 12.0 of 12.0 raised). Yet paper-side recall remains at 0.24. The refiner’s claim of having “addressed” issues appears to be aspirational rather than operational. The current verifier issue types do not include a coverage-gap signal (a fact present in 1–2 of 3 drafts and supported by paper evidence), so the refiner has no positive signal to preserve draft-only specifics.

Mechanism M3 no paper-side awareness in the pipeline itself. The verifier and retriever are issue-driven; the refiner reads only the drafts and verifier output. There is no mechanism that says “*the paper claims X; check that it survives the merge.*” This is hard to falsify directly with the current data, but is suggested by the M1 case studies: facts present in one of three drafts also appear in the paper itself, so the information loss is between drafts and refined output, not between paper and drafts.

What does *not* appear to drive the failure. We had hypothesised that the structured output schema (`tldr` / `core_idea` / `method` / `experiments` / `limitations`) might bias toward abstract phrasing. The data does not support this. B5 LLAMA-70B-STRUCTURED uses essentially the same sectioned prompt as the schema imposes on OURS but writes *single-LLM* drafts. Its F_1 is statistically indistinguishable from B1 Llama-70B-naive (0.405 vs. 0.418, CI overlap), suggesting the structured schema itself is approximately neutral. We retract the structured-schema mechanism we considered initially.

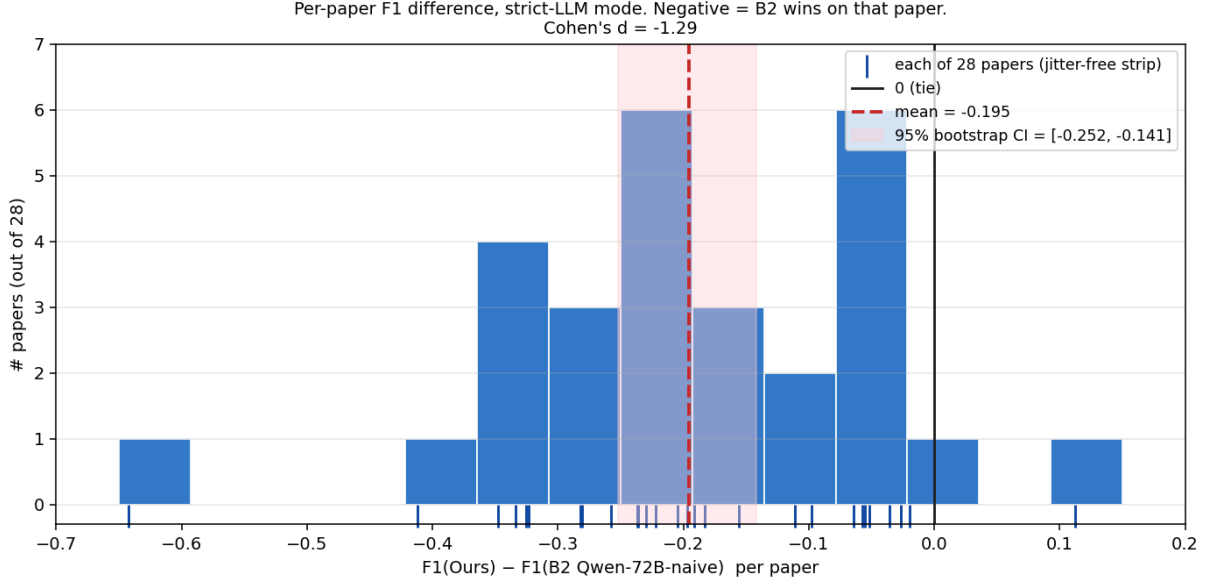


Figure 3: Per-paper F_1 difference $F_1(\text{OURS}) - F_1(\text{B2})$ under strict-LLM. Mean is -0.20 (red dashed). The 95% bootstrap CI $[-0.25, -0.14]$ (red shading) is well to the left of zero.

8 Toward a Coverage-Aware Pipeline

The mechanism candidates point at three actionable fixes (F1, F2, F3). We summarise them here for completeness; **F1** and **F2** are validated as prompt-level interventions in this paper’s appendix material, and **F3**—combined with a non-merging architecture—is validated as a follow-up architectural redesign in the companion paper Sun [2026]. The companion paper’s headline contrast (ONEVISION vs. B6 single-LLM-with-the-same-prompt control) is $\Delta F_1 = +0.106$ strict-LLM, 95% CI $[+0.047, +0.175]$ on $n = 25$ paired papers; $\Delta F_1 = +0.126$ DeBERTa-NLI, CI $[+0.072, +0.187]$, $n = 26$ paired (full $n = 29$ sample). We sketch the three fixes here and defer the full architectural study to that work.

(F1) Union-biased refiner prompt. The v1 prompt’s “merge the strongest content” is replaced with an explicit *preserve-the-union* directive: every specific number, dataset, method name, or quantitative finding present in any draft and supported by evidence must survive the merge. The new prompt explicitly warns against the M1 failure mode (“do not replace ‘28.4 BLEU on WMT 2014 EN-DE’ with ‘state-of-the-art’ because only one draft had the specific number”).

(F2) Verifier reads full paragraph text. The v1 verifier was fed the paragraph *short summaries* (a pre-processing artefact intended to keep verifier prompts compact), which silently filtered out the very specifics the verifier was supposed to flag as missing. The fix is purely a plumbing change: feed the verifier the full paragraph text. This trivial-looking change is, in retrospect, the largest single contributor to the recall deficit we measure in Section 6.

(F3) Paper-aware Step 0 (validated in companion work). A pre-extraction step that produces a list of atomic “must-cover” paper facts before draft generation, conditioning the drafters on this checklist. This is conceptually identical to the paper-claim extractor used by our evaluator (Step 5), and to the key-information-unit [KIU, Anonymous Authors, 2025] concept in iterative legal element extraction. The companion paper Sun [2026] replaces the *merging* refiner with a *vote-then-augment* architecture that selects one winning draft via 3-round game-theoretic voting and then augments it with verifier-flagged missing facts. This combination of F2 (verifier-on-full-text) and F3-like rearchitecting is what recovers the $+0.11$ F_1 result above. Crucially, the

same prompt directives applied to a single-LLM call (B6 in this paper, also B6 in the companion) do *not* reach the same F_1 — settling the prompt-vs.-architecture question raised by this paper’s negative result in favour of *both* mattering, with the architectural contribution conditional on getting the merge step right.

9 Limitations

(i) **Single seed.** We ran the experiment once with seed 42. The three drafters use $T=0.7$, so re-running with a different seed will produce different drafts. The reported within-experiment per-paper std-dev (Table 1: $\sigma_{F_1}=0.15$ for OURS) is a partial proxy for run-to-run noise but does not substitute for multi-seed averaging. That said, the largest gap ($\Delta F_1 = -0.20$ vs. Qwen-72B-naive, $d = -1.29$) is roughly 1.3 within-paper std-dev units; even if multi-seed noise were as large as the within-paper noise, the conclusion that OURS loses to B2 would survive. The conclusion that OURS is statistically *tied* with B1 and B5 is the more fragile claim and could flip either direction with multi-seed averaging.

(ii) **Atomic claims rely on an LLM extractor.** Both Step 1 (summary $\rightarrow$ atomic claims) and Step 5 (paper $\rightarrow$ atomic claims) are produced by Llama-3.3-70B, the same model that participates in our pipeline. Even though the DeBERTa Step-3 verifier removes self-favouring at the verification step, claim *selection* could still favour phrasings preferred by Llama. A definitive test would replace the claim extractor with a non-Llama model.

(iii) **DeBERTa-v3-base context.** The 512-token context truncates long paragraphs. We do not observe systematic bias from this in spot-check.

(iv) **Both judges are pair-judges.** They score one (summary, paper) pair at a time; they cannot diagnose ranking-style errors or relative quality between two summaries directly.

(v) **BM25 retrieval.** A dense retriever could change which evidence is fed to the verifier and shift verdicts.

10 Conclusion

We test whether multi-agent LLM debate, validated as an effective recipe on consensus-style reasoning tasks, transfers to long-form factual summarisation. On 29 arXiv papers, two independent evaluators agree that a 3-LLM debate + RAG-grounded refiner pipeline *loses* to a single-call Qwen-72B baseline by $\Delta F_1 = -0.20$ ($d = -1.29$, 26/28 paper-level losses), driven by paper-side recall: the multi-agent output covers one-fourth as many paper facts despite being the same length. The failure is not uniform: against Llama-70B-family baselines (B1, B5) the gap is statistically indistinguishable from zero. Case studies attribute the recall deficit to a refiner that intersects rather than unions the drafts, supported by an issue-cap-saturated verifier whose self-reported “issues addressed” is suspiciously unanimous.

We frame this as a *task-type* failure: multi-agent debate helps when many agents reduce a single noisy estimate (consensus reasoning), but the version we tested hurts when many agents must aggregate breadth (union reconstruction). The naive transfer between task types is the bug.

Negative result, but not the architectural verdict. Our companion paper Sun [2026] shows that the failure above is not intrinsic to multi-agent debate on union-reconstruction tasks. With two surgical changes — (i) verifier reading full paragraph text instead of paragraph short-summaries, (ii) replacing the merging refiner with a 3-round game-theoretic voter that picks *one* winning draft, then augmenting that draft with verifier-flagged missing facts — the redesigned pipeline beats the same-prompt single-LLM control (B6 in this paper) by $+0.11 F_1$ under both decoupled judges on $n = 25\text{--}26$ paired papers from a 29-paper sample. The negative result we report here is therefore a result about a *specific* multi-agent architecture (debate-as-merge), not about multi-agent debate per se on this task class. We hope the diagnostic case studies and

the paired-CI methodology in this paper help future work avoid the same trap, and that the companion paper’s positive architectural result motivates further variation along the selection-vs.-merge axis.

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